

Optimizing technological configurations in residential energy systems with vehicle-to-home integration and community-level power sharing

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ABSTRACT

Distributed energy technologies, such as photovoltaic (PV), stationary storage batteries (SBs), and vehicle-to-home (V2H) systems, are expected to be introduced in pursuit of carbon neutrality in the residential and transportation sectors. This study developed the model of a two-tiered energy system to optimize technological configurations and examined the impacts of installed capacities of PV and SBs, along with various vehicle types, on the economic and environmental performance, stability, and efficiency of residential energy systems. The primary contribution of this research lies in its comprehensive analysis of energy systems that incorporate bidirectional power supply between vehicles and residences, as well as power sharing among multiple households within a community. The simulation results revealed substantial benefits from the incorporation of V2H systems and SBs, which notably reduced annual energy costs by approximately 30 % and CO₂ emissions by approximately 40 % in scenarios leveraging community electricity sharing, compared to scenarios without such flexibility. This integration significantly enhanced the self-consumption and self-sufficiency rates of residential energy systems by approximately 10 %, illustrating the efficacy of community-level electricity sharing combined with the charging and discharging capabilities of vehicles. This strategy not only optimizes resource use but also reduces dependency on external power supplies, thereby enhancing overall energy efficiency.

1. Introduction

In response to the growing problem of climate change, reducing greenhouse gas emissions has become an important global concern. In Japan, the Prime Minister declared in October 2020 that the country would achieve carbon neutrality by 2050, and the Sixth Strategic Energy Plan, subsequently approved by the Cabinet, set a target of reducing greenhouse gas emissions by 46 % compared to 2013 levels by 2030 (Agency for Natural Resources and Energy, 2021). To achieve carbon neutrality, clean energy vehicles such as hybrid electric vehicles (HEVs), plug-in hybrid electric vehicles (PHEVs), battery electric vehicles (BEVs), and fuel cell vehicles (FCVs) are being developed and introduced to the market in the transportation sector. The residential sector has also begun using distributed energy technologies such as photovoltaic (PV) power generation and stationary storage batteries (SBs). In particular, the recent introduction of a vehicle-to-home (V2H) system using a dedicated power conditioner that connects a PHEV or BEV to the distribution board of the residence is expected to enable a bidirectional power supply between the vehicle and the residence and utilize the vehicle as a residential energy storage system. Because the amount of

electricity generated from renewable energy sources, such as solar power, fluctuates depending on the weather and time of day, resources or means for adjusting electricity supply and demand are needed. The use of batteries in vehicles parked at home is expected to improve energy efficiency and reduce CO₂ emissions in both the residential and transportation sectors.

Numerous studies have evaluated the economic benefits of deploying distributed energy technologies in residential settings. Hoppmann et al. (Hoppmann et al., 2014) and Yu (Yu, 2018) evaluated the cost-effectiveness of PV and SB installations in households in Germany and France, respectively, and analyzed their economic viability. Uddin et al. (Uddin et al., 2017) and Cerino Abidin and Noussan (Cerino Abidin and Noussan, 2018) working in the UK and Italy, respectively, considered the impacts of battery degradation and power losses on the economics of PV and SB installations, noting that these factors could diminish system efficiency and cost-effectiveness. Aniello et al. (Aniello et al., 2021) further analyzed the economic performance of such installations in Germany using various financial indicators, such as payback periods, internal rate of return, and net present value. These studies assumed only SBs as the energy storage technology, but Laurischkat and Jandt (Laurischkat and Jandt, 2018) and Wang et al. (Wang

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Nomenclature	
<i>Abbreviations</i>	
BEVs	Battery electric vehicles
CF	Charging facility
C2H	Community-to-Home
C2S	Community-to-SB
C2V	Community-to-Vehicle
FCVs	Fuel cell vehicles
GVs	Gasoline vehicles
G2H	Grid-to-Home
HEVs	Hybrid electric vehicles
H2C	Home-to-Community
H2G	Home-to-Grid
H2S	Home-to-SB
H2V	Home-to-Vehicle
P/PV	Photovoltaic
PHEVs	Plug-in hybrid electric vehicles
P2H	PV-to-Home (PV power generation)
S/SBs	Stationary storage batteries
S2C	SB-to-Community
S2H	SB-to-Home
V	Vehicle
V2C	Vehicle-to-Community
V2H	Vehicle-to-Home
<i>Symbols</i>	
A^P	Installed area of PV generation [m ²]
AC^P	Upper limit for the installed PV area [m ²]
BC^S	Battery capacity of SBs [kWh]
BC^V	Battery capacity of each vehicle type [kWh]
CE^{annual}	Total CO ₂ emissions [t-CO ₂]
c	Conversion factor [-]
E^{C2H}	Total amount of electricity from community to home [MJ]
$E^{C2H,shortage}$	Amount of electricity from community to home for the shortage of electricity in the household [MJ]
E^{C2V}	Amount of electricity from community to vehicle [MJ]
E^{C2S}	Amount of electricity from community to SB [MJ]
E^{demand}	Household demand for electricity [MJ]
E^{G2H}	Amount of electricity from grid to home [MJ]
$E^{gasoline}$	Amount of gasoline used for driving [MJ]
E^{H2C}	Total amount of electricity from home to community [MJ]
$E^{H2C,surplus}$	Amount of electricity from home to community due to household surpluses [MJ]
E^{H2G}	Amount of electricity from home to grid [MJ]
E^{H2S}	Amount of electricity from home to SB [MJ]
E^{H2V}	Amount of electricity from home to vehicle [MJ]
$E^{hydrogen}$	Amount of hydrogen used for driving [MJ]
E^{P2H}	Amount of electricity from PV to home [MJ]
E^{S2C}	Amount of electricity from SB to community [MJ]
E^{S2H}	Amount of electricity from SB to home [MJ]
E^{V2C}	Amount of electricity from vehicle to community [MJ]
E^{V2H}	Amount of electricity from vehicle to home [MJ]
$E^{V,run}$	Amount of electricity charged outside of the residence (amount of electricity that required additional charging when driving) [MJ]
$G^{net,grid}$	Differences in electricity supply and demand as a precondition for transactions with the grid [MJ]
$G^{net,home}$	Differences in electricity supply and demand in residential homes [MJ]
$I^{gasoline}$	CO ₂ intensity of gasoline [t-CO ₂ /MJ]
I^{grid}	CO ₂ intensity of electricity from the grid [t-CO ₂ /MJ]
$I^{hydrogen}$	CO ₂ intensity of hydrogen [t-CO ₂ /MJ]
N	Number of days [days]
$RC^{S,charge}$	Residual charge capacity of SBs [MJ]
$RC^{S,discharge}$	Residual discharge capacity of SBs [MJ]
$RC^{V,charge}$	Residual charge capacity of vehicle battery [MJ]
$RC^{V,discharge}$	Residual discharge capacity of vehicle battery [MJ]
r	Discount rate [%]
SC^{annual}	Self-consumption rate [%]
SOC^S	State of charge of SBs [kWh]
SOC^V	State of charge of the battery of each vehicle type [kWh]
SS^{annual}	Self-sufficiency rate [%]
T^{CF}	Durable years of charging facility [years]
T^P	Durable years of PV [years]
T^S	Durable years of SBs [years]
T^V	Durable years of vehicle [years]
TC^{annual}	Total annual costs [million yen]
TC^{carbon}	Carbon costs associated with CO ₂ emissions [million yen]
TC^{energy}	Energy costs [million yen]
TC^{tech}	Installation costs for vehicles and equipment [million yen]
UP^{carbon}	Unit price of carbon taxes [million yen/t-CO ₂]
UP^{CF}	Unit price of charging facility [million yen/unit]
UP^{C2H}	Electricity purchase price from community [million yen/MJ]
UP^{G2H}	Electricity purchase price from the grid [million yen/MJ]
$UP^{gasoline}$	Unit price of gasoline [million yen/MJ]
$UP^{hydrogen}$	Unit price of hydrogen [million yen/MJ]
UP^{H2C}	Electricity sale price to community [million yen/MJ]
UP^{H2G}	Electricity sale price to the grid [million yen/MJ]
UP^P	Unit price of PV [million yen/m ²]
UP^S	Unit price of SBs [million yen/kWh]
UP^V	Unit price of each vehicle type [million yen/unit]
<i>Subscripts</i>	
d	Day category [weekday, holiday]
h	Household type [1–5]
k	Target year [2030]
m	Month [1–12]
t	Hour [0–23]
w	Weather category [sunny, cloudy, other]

et al., 2020) investigated the economic feasibility of installing PV, SBs, and BEVs in households in Germany and the United Kingdom, respectively. Chen et al. (Chen et al., 2020) created four cases for households in Shanghai, China, with and without PV and with BEV charge/discharge patterns (charge only or both charge and discharge) and evaluated their economic potential. The results demonstrated that V2H significantly enhances the utilization of PV-generated power, yielding greater economic benefits and more efficient energy use than merely transferring valley electricity. With the goal of expanding the scope of renewable energy solutions in residential energy systems, García-Vázquez et al. (García-Vázquez et al., 2022) assessed the feasibility of integrating small

wind turbines with PV, SBs, and BEVs in homes in Spain.

In addition to economic considerations, some studies have examined the environmental impacts of distributed energy technologies. Kobashi et al. (Kobashi et al., 2020; Kobashi and Yurime, 2019) considered a cost decline for PV, SBs, and BEVs through 2030 in Kyoto, Japan, and Shenzhen, China. They assessed the economic and environmental benefits of combining PV with V2H systems in comparison to only PV and SBs systems in future scenarios in Japan. Osawa et al. (Osawa et al., 2019) analyzed the self-sufficiency and self-consumption rates, along with the CO₂ emissions, of various configurations, including gasoline vehicles (GVs), BEVs, GV with SBs, BEV with SBs, and BEV with V2H

systems. They found that integrating BEVs with V2H systems could potentially reduce CO₂ emissions more than BEVs alone. Akimoto et al. (Akimoto et al., 2020) investigated operational methods for using surplus PV electricity, such as charging SBs, charging BEVs, and operating heat-pump water heaters during the day. Their findings indicated that households with infrequent vehicle use and short driving distances achieved higher self-consumption of surplus PV power, resulting in lower CO₂ emissions and annual costs. In contrast, households with frequent vehicle use and longer driving distances were less effective at utilizing surplus PV power. Dewi et al. (Dewi et al., 2023) analyzed the effects of introducing PV and BEVs in Jakarta, Indonesia, and revealed the potential for reducing energy system costs and CO₂ emissions at the city level.

Furthermore, several studies have focused on optimizing equipment configurations and operating schedules in residences to enhance the efficiency of residential energy systems. Higashitani et al. (Higashitani et al., 2022, 2021) optimized configurations and operations of SBs and gas water heaters to minimize annual costs in households with BEVs and combined BEVs with V2H systems, finding economic benefits in integrating V2H systems and SBs especially when BEVs are frequently absent. Additionally, new installations of heat-pump water heaters were proven to be cost-effective. Erdinc et al. (Erdinc et al., 2015) developed a model to optimize PV and SB capacities for residences with a BEV and V2H system. Wu et al. (Wu et al., 2017) focused on optimizing SB size and battery operation strategies in households with PVs and BEVs and noted that V2H systems could significantly reduce daily electricity costs. Doroudchi et al. (Doroudchi et al., 2018) explored heating system scenarios in homes with PV and BEVs, with the goal of minimizing grid electricity use by leveraging SBs and V2H technology. Naghibi et al. (Naghibi et al., 2018) assessed cost-minimizing configurations of PV, wind turbines, and SBs in households utilizing V2H systems with BEVs. Furthermore, Wu et al. (Wu et al., 2016) and Lazzaroni et al. (Lazzaroni et al., 2019) devised a charge and discharge schedule for BEVs that minimizes electricity costs in households with PV, considering uncertainties in vehicle usage patterns. Niu et al. (Niu et al., 2024) constructed a stochastic model to simulate random driving patterns and optimized the household energy system's configuration, including decisions on whether to include a V2H system, as well as the capacities of SBs and PVs, to minimize equivalent annual costs. The study then analyzed how BEV adoption and stochastic mobility behavior influenced energy costs, CO₂ emissions, and energy self-sufficiency. Considering the benefits and opportunities of free charging at workplaces and public areas, Wu et al. (Wu et al., 2024) analyzed the capacity of PV systems and SBs to minimize annual energy costs for households using V2H systems. Gudmunds et al. (Gudmunds et al., 2020) investigated the optimal sizes of PV and SBs to reduce external BEV charging, and Abdalla et al. (Abdalla et al., 2023) optimized BEV and SB operating schedules, focusing on maximizing PV electricity self-consumption alongside cost efficiency. Subsequently, Osawa (Osawa, 2024) developed a residential energy system model incorporating a bidirectional power flow between residence and vehicle that optimizes vehicle types and sizes of PV and SBs to minimize both annual costs and CO₂ emissions. Zhu et al. (Zhu et al., 2023) investigated optimal charge and discharge schedules for BEVs and SBs to minimize energy costs and CO₂ emissions in homes equipped with PV, SB, and BEV systems under different electricity price scenarios in China and Germany. Although V2H systems were not considered, several studies have analyzed the effects of power sharing among multiple residences. Mota et al. (Mota et al., 2024) focused on minimizing energy costs by shifting home appliance loads to periods when energy prices are lower, taking into account PV generation and dynamic energy pricing. Their study also compared the impact on energy costs when considering a single household versus multiple households engaging in energy sharing. Similarly, Jeddi et al. (Jeddi et al., 2021) proposed a distributed load scheduling framework to coordinate the operation of multiple household energy management systems within a residential neighborhood.

They analyzed the electricity consumption schedules of household appliances and the charge and discharge schedules of SBs to minimize the electricity payments for individual households.

Previous studies have evaluated the effectiveness and operational strategies of BEVs and V2H systems from multiple perspectives and have provided valuable insights. Although previous research has explored the sizing and operating schedules of distributed energy technologies within individual homes, assuming various electricity consumption patterns, there has been insufficient consideration of power sharing within a community containing multiple households. While some studies (Jeddi et al., 2021; Mota et al., 2024) have analyzed electricity sharing between multiple residences, they did not consider bidirectional power supply between residences and vehicles. Strategic exchange of electricity among households with different PV capacities, electricity consumption patterns, and vehicle usage schedules has the potential to significantly increase energy self-sufficiency. Additionally, although scholars (Abdalla et al., 2023; Osawa, 2024; Zhu et al., 2023) have applied multiple objective functions to optimize residential energy systems, there remains a gap in comprehensive studies that integrate analyses of economic performance, environmental performance, stability, and efficiency. Furthermore, most existing studies have focused primarily on BEVs and have not considered other vehicle types such as HEVs, PHEVs, and FCVs as design variables in the optimization model. Because different powertrains are being developed with the goal of achieving carbon neutrality and satisfying diverse consumer preferences, it is important to consider the characteristics of these various vehicle types when designing future energy systems. To the best of our knowledge, there is a significant research gap in comprehensively analyzing energy systems at the community level, where multiple households are considered, and bidirectional power supply between residences and vehicles is incorporated, as shown in Table 1.

To fill the identified research gaps, this study aimed to analyze the impacts of installed capacities of PV and SBs, various vehicle types, and their combinations on the economic performance, environmental performance, stability, and efficiency of residential energy systems. This analysis considered the bidirectional power supply between vehicles and residences, as well as power sharing within the community. To achieve this objective, the study assumed a two-tiered energy system consisting of an individual residence and a community containing multiple residences and developed a model to evaluate and optimize the technological configurations. By separating the optimization into two layers (two steps), the problem could be simplified, the behavior of the entire system could be understood more fully, and the computational complexity could be reduced. The main contributions of this study follow:

- Effective technological configurations and their sizing in residential energy systems are clarified.
- The effectiveness of a bidirectional power supply between vehicles and residences, along with community-level power sharing, is evaluated to determine its impact on the economic, environmental, stability, and efficiency outcomes of energy systems.
- A two-tier optimization model that considers the interests of both households and the community is introduced, and a comprehensive framework oriented toward total system optimization is proposed.

Such information will be valuable for guiding the design of residential energy systems and local microgrids for more efficient energy use and for developing future scenarios that promote the diffusion of distributed energy technologies. Section 2 of this paper describes the new model for evaluating and optimizing the technological configuration of residential energy systems. Section 3 elaborates on the various prerequisite data for the model. Section 4 presents and explains the simulation results generated by the model, and Section 5 presents the conclusions of this study and discusses future prospects based on the discoveries made herein.

Table 1
Comparison with previous studies on the optimization of residential energy systems.

References	Objective function	Boundary	Features considered in the optimization			
			Vehicle types	V2H	Power sharing	Thermal energy
(Higashitani et al., 2022, 2021)	Annual costs	Individual households	GV, BEV	✓		✓
(Erdinc et al., 2015)	Total net present value of cash flows	Individual households	BEV	✓		
(Wu et al., 2017)	Total costs of electricity and battery energy storage system	Individual households	BEV	✓		
(Doroudchi et al., 2018)	Grid electricity use	Individual households	BEV	✓		✓
(Naghibi et al., 2018)	Electricity costs	Individual households	BEV	✓		
(Wu et al., 2016)	Electricity costs	Individual households	BEV	✓		
(Lazzeroni et al., 2019)	Electricity costs	Individual households	BEV	✓		✓
(Niu et al., 2024)	Annual costs	Individual households	GV, BEV	✓		✓
(Wu et al., 2024)	Energy costs	Individual households, External locations (Workplace and public charging)	BEV	✓		
(Gudmunds et al., 2020)	External BEV charging	Individual households	BEV	✓		
(Abdalla et al., 2023)	Cost efficiency, PV electricity self-consumption	Individual households	BEV	✓		
(Osawa, 2024)	Annual costs, CO ₂ emissions	Individual households	GV, HEV, PHEV, BEV, FCV	✓		
(Zhu et al., 2023)	Energy costs, CO ₂ emissions	Individual households	BEV	✓		
(Mota et al., 2024)	Energy costs	Individual households, Community (Multiple households)	-		✓	
(Jeddi et al., 2021)	Electricity payments	Community (Multiple households)	-		✓	
This study	Annual costs, CO ₂ emissions, Self-consumption rate, Self-sufficiency rate	Individual households, Community (Multiple households)	GV, HEV, PHEV, BEV, FCV	✓	✓	

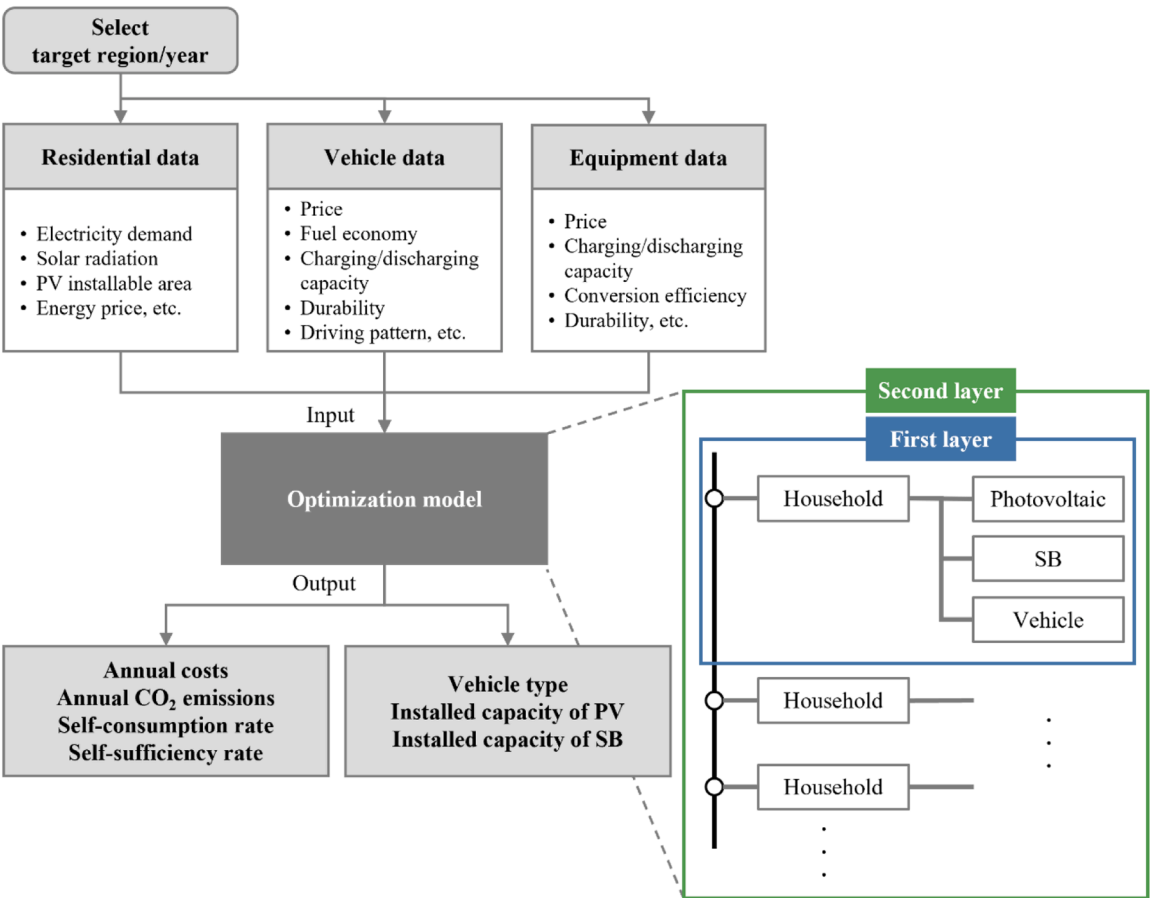


Fig. 1. Overview of the analytical model.

2. Methods

2.1. Framework

An overview of the analytical model constructed in this study is shown in Fig. 1. This model incorporates three input categories: residential data, vehicle data, and equipment data. The first step is to select the target region and target year. The target region and year selected for this study, as a case study, were the Kanto region of Japan and 2030. Based on the selected target region, residential data such as electricity demand, solar radiation, and electricity prices were specified. The model operates with a time resolution of one hour, allowing it to capture fluctuations in electricity demand, grid electricity prices, and CO₂ intensity, all of which vary hourly. Notably, grid electricity prices and CO₂ intensity typically decrease during daytime hours, in coincidence with increased PV generation, a trend reflected in our model. Additionally, the model integrates vehicle and equipment data, which include parameters such as price, charge/discharge capacity, and durability years. Given the important impact of vehicle absence time on the economic and environmental performance, as well as the efficiency of V2H systems, the model considers multiple driving patterns to assess these effects robustly. Based on these inputs, an optimization model was developed to derive the optimal technological configurations by calculating the charging and discharging of energy storage technologies, annual costs, annual CO₂ emissions, self-consumption rate, and self-sufficiency rate.

This optimization model assumed a two-tier energy system to address the interests of both individual households and a community consisting of multiple households. Initially, in the first layer, Pareto-optimal technological configurations were calculated for each household alone. These configurations considered vehicle type, PV capacity, SB capacity, and the inclusion or exclusion of the V2H system, with the dual objectives of minimizing annual costs and CO₂ emissions. We then normalized the objective function values of the solutions using the Min–Max normalization method. To accommodate diverse consumer preferences and examine potential trade-offs between the two objectives, we implemented five distinct weightings in 0.25 increments across a 0–1 scale. The specific weight pairs applied to annual costs and CO₂ emissions were 1 and 0, 0.75 and 0.25, 0.5 and 0.5, 0.25 and 0.75, and 0 and 1, respectively. For each weighting scenario, the highest combined value of the weighted objective functions was used to select five optimal solutions for each household. Subsequently, at the second layer, we calculated the Pareto-optimal technological configurations for the community as a whole based on the solutions selected in the first layer, employing four objective functions: minimizing annual costs and annual CO₂ emissions while maximizing self-consumption and self-sufficiency rates, with consideration of the electricity sharing among multiple households within the community. By structuring the optimization as two sequential layers, the problem was simplified, and the complexity of the potential combinations was mitigated, effectively aligning the interests of individual households with community objectives. This layered approach facilitates a multifaceted analysis critical for informed decision-making in addressing complex environmental and energy challenges, involving multiple stakeholders. In this study, multi-objective optimization was performed using genetic algorithms of Pymoo (Blank and Deb, 2020). The mathematical formulae of the optimization model are described in the following sections. A genetic algorithm is an optimization method that emulates the process of natural selection, iteratively evolving a population of potential solutions towards an optimal set, known as the Pareto frontier. This method is particularly effective for the simultaneous optimization of multiple objective functions, as it performs a multi-point search across the entire solution space, enabling a comprehensive exploration of potential solutions. By applying genetic algorithms, the model efficiently navigates the trade-offs between multiple objectives in residential energy systems, allowing for the identification of solutions that balance economic performance, environmental performance, stability, and efficiency.

A flow chart depicting the residential energy system within the analytical model of this study is illustrated in Fig. 2. The model addresses both the electricity and vehicle driving demands of individual households, as well as the electricity shortages and surpluses within the community. In the first layer of analysis, the model focuses solely on the household demands for electricity and vehicle usage. The second layer expands the focus to include community-wide electricity shortages and surpluses. The model incorporated a variety of vehicle types and various PV and SB capacities, thus allowing for dynamic analyses of electricity demand and supply adjustments for each configuration. It operates with an hourly time resolution and simulates operations across 72 representative days categorized by month, weekday/holiday, and weather condition (sunny, cloudy, or other). These days are further multiplied by the number of days in each category for each region to calculate values for one year. Regarding vehicle integration, the model considered currently market-available vehicle types compatible with the V2H system, specifically PHEVs and BEVs. It assumed that bidirectional power supply to residences from these vehicles is possible only when the vehicles are present at each residence.

2.2. Optimization model

2.2.1. First layer: individual household level

The optimization model of this study incorporated two primary objective functions at the first layer: annual costs and annual CO₂ emissions, as detailed in Eqs. (1) and (2), respectively. To minimize these objective functions for each household, the model calculates the optimal vehicle type, PV installed area, SB installed capacity, and whether to include a V2H system. The annual costs are subdivided into three main categories: installation costs for vehicles and equipment, energy costs, and carbon costs associated with CO₂ emissions, as outlined in Eq. (1). Specifically, the installation costs include expenditures for the charging facilities necessary for home charging of BEVs and PHEVs. To facilitate financial analysis over time, the installation costs of vehicles and equipment, which are fixed costs, were annualized using the durable years of the assets and an interest rate of 3 %, as formulated in Eq. (3). The energy costs aggregate expenditures on electricity, gasoline, and hydrogen, as presented in Eqs. (1) and (4). Here, the electricity cost calculation considers both the expenses incurred from grid power purchases and the revenues from selling surplus power generated by PV systems. The model operates with an hourly time resolution and evaluates a total of 72 representative days. These days are categorized by month, weekday/holiday status, and weather condition (sunny, cloudy, or other). The calculations extend these daily values to an annual scale by multiplying them by the number of days in each category. Moreover, many countries are adopting or contemplating carbon taxes to achieve carbon neutrality. In Japan, such a tax is partially implemented as a “tax for global warming countermeasures” (Ministry of the Environment, 2012). In our model, the applicable carbon tax rate is applied to the CO₂ emissions resulting from each technological configuration, and this cost is incorporated into the annual costs, presented in Eqs. (1) and (5). Annual CO₂ emissions are calculated by multiplying the annual purchases of grid power, gasoline, and hydrogen by their respective CO₂ intensities, as specified in Eq. (2). It is important to note that in our calculations, the CO₂ emissions offset by the sale of surplus electricity generated by PV systems to the grid are not considered a reduction in emissions. Furthermore, Eq. (6) delineates the energy balance, ensuring that hourly residential electricity demand and supply are aligned. The system is designed to manage residential electricity consumption, PV generation, and vehicle and SB charging/discharging. Specifically, it purchases electricity from the grid when demand exceeds supply and sells electricity back to the grid when supply exceeds demand, as delineated in Eqs. (7) and (8). Moreover, based on previous research (Osawa, 2024; Sugihara et al., 2011), hourly PV generation is calculated by multiplying the amount of solar radiation by the installed area, module conversion efficiency, temperature loss

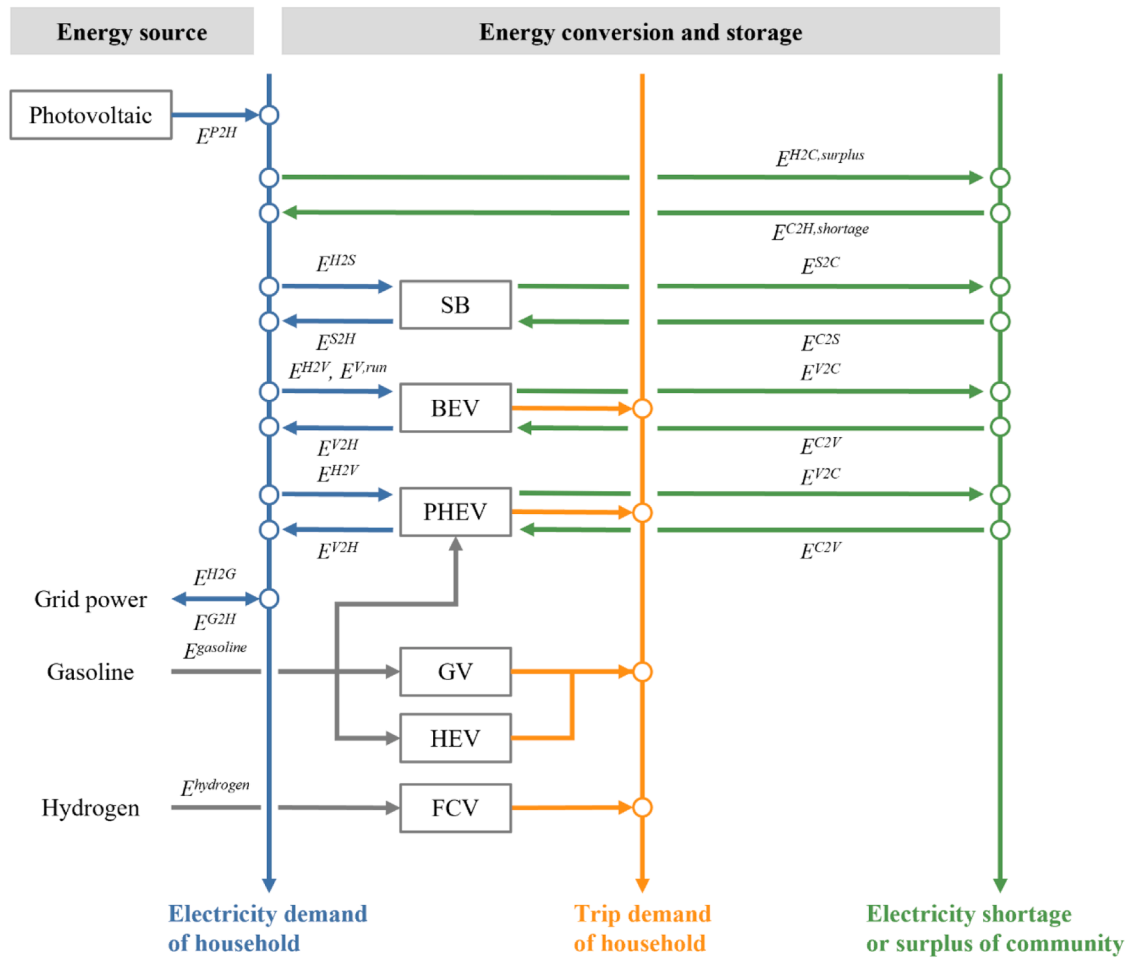


Fig. 2. Flow chart of the modeled residential energy system.

factor, and integration factor.

$$\min TC_{hk}^{annual} = TC_{hk}^{tech} + \sum_{m,d,w} N_{k,m,d,w} (TC_{hk,m,d,w}^{energy} + TC_{hk,m,d,w}^{carbon}) \quad (1)$$

$$\min CE_{hk}^{annual} = \sum_{t,m,d,w} N_{k,t,m,d,w} (E_{h,k,t,m,d,w}^{G2H} I_{t,m,d,w}^{grid} + E_{h,k,t,m,d,w}^{gasoline} I_k^{gasoline} + E_{h,k,t,m,d,w}^{hydrogen} I_k^{hydrogen}) \quad (2)$$

$$TC_{hk}^{tech} = UP_{hk}^V \frac{r(1+r)^{T^V}}{(1+r)^{T^V} - 1} + UP_{hk}^{CF} \frac{r(1+r)^{T^{CF}}}{(1+r)^{T^{CF}} - 1} + A_{hk}^P UP_k^P \frac{r(1+r)^{T^P}}{(1+r)^{T^P} - 1} + BC_{hk}^S UP_{hk}^S \frac{r(1+r)^{T^S}}{(1+r)^{T^S} - 1} \quad (3)$$

$$TC_{hk,m,d,w}^{energy} = \sum_t (E_{h,k,t,m,d,w}^{G2H} UP_{t,m,d,w}^{G2H} - E_{h,k,t,m,d,w}^{H2G} UP_{t,m,d,w}^{H2G} + E_{h,k,t,m,d,w}^{gasoline} UP_k^{gasoline} + E_{h,k,t,m,d,w}^{hydrogen} UP_k^{hydrogen}) \quad (4)$$

$$TC_{hk,m,d,w}^{carbon} = UP_k^{carbon} \sum_t (E_{h,k,t,m,d,w}^{G2H} I_{t,m,d,w}^{grid} + E_{h,k,t,m,d,w}^{gasoline} I_k^{gasoline} + E_{h,k,t,m,d,w}^{hydrogen} I_k^{hydrogen}) \quad (5)$$

$$G_{h,k,t,m,d,w}^{net,grid} = (E_{h,k,t,m,d,w}^{demand} + E_{h,k,t,m,d,w}^{H2V} + E_{h,k,t,m,d,w}^{H2S}) - (E_{h,k,t,m,d,w}^{P2H} + E_{h,k,t,m,d,w}^{V2H} + E_{h,k,t,m,d,w}^{S2H}) \quad (6)$$

$$E_{h,k,t,m,d,w}^{G2H} = \begin{cases} G_{h,k,t,m,d,w}^{net,grid}, & \text{if } G_{h,k,t,m,d,w}^{net,grid} \geq 0 \\ 0, & \text{otherwise} \end{cases} \quad (7)$$

$$E_{h,k,t,m,d,w}^{H2G} = \begin{cases} 0, & \text{if } G_{h,k,t,m,d,w}^{net,grid} \geq 0 \\ -G_{h,k,t,m,d,w}^{net,grid}, & \text{otherwise} \end{cases} \quad (8)$$

In this model, operational constraints were established for the V2H and SB systems. Firstly, electricity generated by PV panels is primarily used for residential consumption. Any surplus electricity is then directed to charge either the SB or the vehicle, depending on the system's configuration. If a surplus remains after charging the SB or vehicle, it is sold to the grid. Conversely, if residential consumption exceeds the electricity generated by the PV, the deficit is covered first by discharging from the SB or vehicle. If a shortage persists even after utilizing the stored energy, additional electricity is purchased from the grid. A specific hierarchy is observed when both V2H systems and SBs are operational: the V2H system is given priority. This means that after the vehicle has been charged or discharged via the V2H system, any remaining surplus or deficit is managed by the SB. Furthermore, the model restricts simultaneous transactions in purchasing electricity from the grid and selling it to the grid, as well as simultaneous charging and discharging of SBs and vehicles. The model supports two types of vehicles capable of

Table 2
Technology parameters.

Item	Value
Vehicle	Price [million yen]
	Approximately 1.8 (GV), 2.2 (HEV), 2.7 (PHEV), 2.8 (BEV), 4.1 (FCV)
	Durability [year]
	14
	Charging capacity [kWh]
	6 (V2H), 3 (Others)
Charging facility for PHEVs and BEVs	Discharging capacity [kWh]
	6 (V2H)
	Lower limit rate of discharge [%]
	40
	Rate of initial SOC [%]
	50
SB	Price [million yen]
	Approximately 0.096
PV	Durability [year]
	10
	Price [million yen/kWh]
	0.042
	Durability [year]
	15
	Storage capacity [kWh]
	0, 4, 6, 8, 10, 12
	Charging and discharging capacity [kWh]
	2.5
PV	Lower limit rate of discharge [%]
	20
	Rate of initial SOC [%]
	50
	Price [million yen/kW]
	Approximately 0.18
	Durability [year]
	25
PV	Installation area [m ²]
	0, 6, 11, 16, 21, 26, 31, 36, 41, 46, 51
	Maximum installable area (Kanto region) [m ²]
	43
	Module conversion efficiency [-]
	0.18
PV	Temperature loss factor [-]
	0.9 (December–February), 0.8 (June–August), 0.85 (Others)
PV	Integration factor [-]
	0.7

bidirectional energy transactions with residences: PHEVs and BEVs. These vehicles are designed to engage in charging and discharging processes only when parked at the residence. Furthermore, the electricity stored in the batteries of SBs, PHEVs, and BEVs must not exceed the respective storage capacities of the batteries, as specified in Eqs. (9) and (10). If the battery charge level decreases below a predetermined lower limit at the start of any hour, the system automatically initiates charging to restore the battery's charge. Notably, during driving, BEVs and PHEVs have no lower discharge limit, allowing for uninterrupted vehicle operation. However, if the charge level falls below zero, the vehicles are programmed to recharge externally. Drawing on findings from previous studies (Pevet et al., 2020; Ueda and Ota, 2022), the lower limit for charge in BEVs and PHEVs was set at 40 % of their storage capacity to ensure mobility; for SBs, it was set at 20 % to provide a reserve for emergency use. The net amount of electricity entering or leaving the storage battery each hour directly corresponds to the changes in the state of charge (SOC) of the battery, as quantified in Eqs. (11) and (12). Furthermore, the area available for PV installations is constrained by the size of detached houses, which varies across different prefectures. To determine the installable PV area in the Kanto region, the average area of detached houses in prefectures within the region—including Ibaraki, Tochigi, Gunma, Saitama, Chiba, Tokyo, Kanagawa, and Yamanashi—was multiplied by a predefined installable area ratio. This product then established the maximum allowable area for PV installation. The calculated values were incorporated as the upper limits for the PV installation area, as shown in Eq. (13).

$$SOC_{h,k,t,m,d,w}^V \leq BC_{h,k}^V \quad (9)$$

$$SOC_{h,k,t,m,d,w}^S \leq BC_{h,k}^S \quad (10)$$

$$SOC_{h,k,t,m,d,w}^V = SOC_{h,k,t-1,m,d,w}^V + c \left(E_{h,k,t,m,d,w}^{H2V} - E_{h,k,t,m,d,w}^{V2H} - E_{h,k,t,m,d,w}^{V,run} \right) \quad (11)$$

$$SOC_{h,k,t,m,d,w}^S = SOC_{h,k,t-1,m,d,w}^S + c \left(E_{h,k,t,m,d,w}^{H2S} - E_{h,k,t,m,d,w}^{S2H} \right) \quad (12)$$

$$A_{h,k}^P \leq AC_h^P \quad (13)$$

2.2.2. Second layer: community level

In the second layer of the optimization model, four objective functions were integrated at the community level: annual costs, annual CO₂ emissions, PV self-consumption rate, and energy self-sufficiency rate, as described in Eqs. (14) to (17). This layer of the model is designed to determine the optimal vehicle type, area for PV installation, capacity of an SB, and inclusion of a V2H system for each household. It also manages the flexibility of electricity distribution among households within the community. Initially, the model calculates the hourly electricity supply–demand gap for each household, as detailed in Eq. (18), facilitating electricity transactions from households with surplus PV-generated electricity to those experiencing shortages. Subsequent transactions involve charging and discharging from vehicles and SBs according to the residual capacities available, as outlined in Eqs. (19) to (22). This approach prioritizes the use of V2H systems over SBs when both are operational, ensuring that electricity transactions within the community optimize the existing infrastructure. Furthermore, the model imposes constraints on the simultaneous charging and discharging of vehicles and SBs to prevent conflicts in energy management. For instance, if a vehicle or SB is being charged due to a surplus of PV electricity in a household, it can only engage in charging activities with the community. Finally, any remaining surpluses or shortages of electricity within the community are addressed by selling to or buying from the grid, as specified in Eq. (26). Consistent with the first layer, the net electricity flow in and out of the storage batteries directly correlates with changes in the SOC, as quantified in Eqs. (28) and (29), ensuring a comprehensive and dynamic management of energy resources within the community.

$$\min TC_k^{annual} = \sum_h \left[TC_{h,k}^{tech} + \sum_{m,d,w} N_{k,m,d,w} \left(TC_{h,k,m,d,w}^{energy} + TC_{h,k,m,d,w}^{carbon} \right) \right] \quad (14)$$

$$\min CE_k^{annual} = \sum_h \sum_{t,m,d,w} N_{k,m,d,w} \left(E_{h,k,t,m,d,w}^{G2H} + E_{h,k,t,m,d,w}^{grid} + E_{h,k,t,m,d,w}^{gasoline} + E_{h,k,t,m,d,w}^{hydrogen} \right) \quad (15)$$

$$\min \{ (1 - SC_k^{annual}), (1 - SS_k^{annual}) \}$$

$$SC_k^{annual} = \frac{\sum_h \sum_{t,m,d,w} \left(E_{h,k,t,m,d,w}^{P2H} - E_{h,k,t,m,d,w}^{H2G} \right)}{\sum_h \sum_{t,m,d,w} E_{h,k,t,m,d,w}^{P2H}} \quad (16)$$

$$SS_k^{annual} = \frac{\sum_h \sum_{t,m,d,w} \left(E_{h,k,t,m,d,w}^{P2H} - E_{h,k,t,m,d,w}^{H2G} \right)}{\sum_h \sum_{t,m,d,w} \left(E_{h,k,t,m,d,w}^{G2H} + E_{h,k,t,m,d,w}^{gasoline} + E_{h,k,t,m,d,w}^{hydrogen} + E_{h,k,t,m,d,w}^{P2H} - E_{h,k,t,m,d,w}^{H2G} \right)} \quad (17)$$

$$G_{h,k,t,m,d,w}^{net,home} = \left(E_{h,k,t,m,d,w}^{demand} + E_{h,k,t,m,d,w}^{H2V} + E_{h,k,t,m,d,w}^{H2S} \right) - \left(E_{h,k,t,m,d,w}^{P2H} + E_{h,k,t,m,d,w}^{V2H} + E_{h,k,t,m,d,w}^{S2H} \right) \quad (18)$$

$$E_{h,k,t,m,d,w}^{V2C} = \begin{cases} RC_{h,k,t,m,d,w}^{V,discharge}, & \text{if } \sum_h G_{h,k,t,m,d,w}^{net,home} \geq \sum_h RC_{h,k,t,m,d,w}^{V,discharge} \\ \sum_h G_{h,k,t,m,d,w}^{net,home} \frac{RC_{h,k,t,m,d,w}^{V,discharge}}{\sum_h RC_{h,k,t,m,d,w}^{V,discharge}}, & \text{if } 0 < \sum_h G_{h,k,t,m,d,w}^{net,home} < \sum_h RC_{h,k,t,m,d,w}^{V,discharge} \end{cases} \quad (19)$$

technological configurations on effective energy management strategies. The first scenario, termed the variable scenario, features variable electricity prices from the grid that fluctuate based on the time of day. In contrast, the fix scenario operates under a fixed electricity price throughout the day. This structure allows us to examine how price fluctuations influence energy management practices within individual households and across the community. The second scenario examines the role of PHEVs and BEVs within the community's energy management system. This includes a charge and discharge scenario, by which

$$E_{h,k,t,m,d,w}^{C2V} = \begin{cases} RC_{h,k,t,m,d,w}^{V,charge}, & \text{if } -\sum_h G_{h,k,t,m,d,w}^{net,home} \geq \sum_h RC_{h,k,t,m,d,w}^{V,charge} \\ -\sum_h G_{h,k,t,m,d,w}^{net,home} \frac{RC_{h,k,t,m,d,w}^{V,charge}}{\sum_h RC_{h,k,t,m,d,w}^{V,charge}}, & \text{if } 0 < -\sum_h G_{h,k,t,m,d,w}^{net,home} < \sum_h RC_{h,k,t,m,d,w}^{V,charge} \end{cases} \quad (20)$$

$$E_{h,k,t,m,d,w}^{S2C} = \begin{cases} RC_{h,k,t,m,d,w}^{S,discharge}, & \text{if } \left(\sum_h G_{h,k,t,m,d,w}^{net,home} - \sum_h RC_{h,k,t,m,d,w}^{V,discharge} \right) \geq \sum_h RC_{h,k,t,m,d,w}^{S,discharge} \\ \left(\sum_h G_{h,k,t,m,d,w}^{net,home} - \sum_h RC_{h,k,t,m,d,w}^{V,discharge} \right) \frac{RC_{h,k,t,m,d,w}^{S,discharge}}{\sum_h RC_{h,k,t,m,d,w}^{S,discharge}}, & \text{if } 0 < \left(\sum_h G_{h,k,t,m,d,w}^{net,home} - \sum_h RC_{h,k,t,m,d,w}^{V,discharge} \right) < \sum_h RC_{h,k,t,m,d,w}^{S,discharge} \end{cases} \quad (21)$$

$$E_{h,k,t,m,d,w}^{C2S} = \begin{cases} RC_{h,k,t,m,d,w}^{S,charge}, & \text{if } \left(-\sum_h G_{h,k,t,m,d,w}^{net,home} - \sum_h RC_{h,k,t,m,d,w}^{V,charge} \right) \geq \sum_h RC_{h,k,t,m,d,w}^{S,charge} \\ \left(-\sum_h G_{h,k,t,m,d,w}^{net,home} - \sum_h RC_{h,k,t,m,d,w}^{V,charge} \right) \frac{RC_{h,k,t,m,d,w}^{S,charge}}{\sum_h RC_{h,k,t,m,d,w}^{S,charge}}, & \text{if } 0 < \left(-\sum_h G_{h,k,t,m,d,w}^{net,home} - \sum_h RC_{h,k,t,m,d,w}^{V,charge} \right) < \sum_h RC_{h,k,t,m,d,w}^{S,charge} \end{cases} \quad (22)$$

$$TC_{h,k,m,d,w}^{energy} = \sum_t \left(E_{h,k,t,m,d,w}^{G2H} UP_{t,m,d,w}^{G2H} - E_{h,k,t,m,d,w}^{H2G} UP_{t,m,d,w}^{H2G} + E_{h,k,t,m,d,w}^{C2H} UP_{t,m,d,w}^{C2H} - E_{h,k,t,m,d,w}^{H2C} UP_{t,m,d,w}^{H2C} + E_{h,k,t,m,d,w}^{gasoline} UP_k^{gasoline} + E_{h,k,t,m,d,w}^{hydrogen} UP_k^{hydrogen} \right) \quad (23)$$

$$E_{h,k,t,m,d,w}^{C2H} = E_{h,k,t,m,d,w}^{C2H,shortage} + E_{h,k,t,m,d,w}^{C2V} + E_{h,k,t,m,d,w}^{C2S} \quad (24)$$

$$E_{h,k,t,m,d,w}^{H2C} = E_{h,k,t,m,d,w}^{H2C,surplus} + E_{h,k,t,m,d,w}^{V2C} + E_{h,k,t,m,d,w}^{S2C} \quad (25)$$

$$G_{h,k,t,m,d,w}^{net,grid} = G_{h,k,t,m,d,w}^{net,home} - E_{h,k,t,m,d,w}^{C2H,shortage} + E_{h,k,t,m,d,w}^{H2C,surplus} \quad (26)$$

$$\sum_h E_{h,k,t,m,d,w}^{C2H} = \sum_h E_{h,k,t,m,d,w}^{H2C} \quad (27)$$

vehicles can both absorb surplus power from and supply power back to the community, depending on the household's energy balance. The alternative charge only scenario restricts vehicles to merely charging, prohibiting any energy discharge back to the community, irrespective of the household's energy surplus or shortage. The Base Scenario, which combines the variable scenario with the charge and discharge scenario, serves as our primary experimental condition. Conversely, the Reference Scenario, which includes the fix scenario and the charge only scenario, is used as a comparative benchmark to illustrate the effects of a more static energy pricing and management approach.

These scenarios are designed to assess how different electricity pricing and vehicle integration strategies can impact the sustainability and cost-effectiveness of community energy systems. By studying these interactions, our research seeks to enhance understanding of optimal energy management practices and contribute to the broader discourse

$$SOC_{h,k,t,m,d,w}^V = SOC_{h,k,t-1,m,d,w}^V + c \left(E_{h,k,t,m,d,w}^{H2V} - E_{h,k,t,m,d,w}^{V2H} - E_{h,k,t,m,d,w}^{V,run} + E_{h,k,t,m,d,w}^{C2V} - E_{h,k,t,m,d,w}^{V2C} \right) \quad (28)$$

$$SOC_{h,k,t,m,d,w}^S = SOC_{h,k,t-1,m,d,w}^S + c \left(E_{h,k,t,m,d,w}^{H2S} - E_{h,k,t,m,d,w}^{S2H} + E_{h,k,t,m,d,w}^{C2S} - E_{h,k,t,m,d,w}^{S2C} \right) \quad (29)$$

2.3. Scenarios

In this study, we developed two main scenarios to explore the economic, environmental, stability, and efficiency impacts of various

on integrating renewable energy sources and smart grid technologies in urban settings.

3. Prerequisite data

3.1. Technology parameters

In this study, the parameter values for vehicles and equipment, as listed in Table 2, were established based on prior studies (Higashitani et al., 2021; Kobashi et al., 2020; Osawa, 2024, 2023). As distributed energy technologies are in the early stages of market penetration and have initially high costs, prices for specific vehicle types, charging facility, and PV were set according to projected future values, reflecting an expected decrease in costs. Previous studies (Pevce et al., 2020; Ueda and Ota, 2022) have indicated that many users tend to recharge BEVs when their SOC drops to approximately 30–35 % due to range anxiety concerns; thus, we set 40 % as the minimum SOC threshold for discharge purposes. For SBs, a lower limit discharge rate of 20 % was designated to accommodate emergency scenarios. The optimal sizes for PV installation areas and SB capacities are anticipated to vary based on the technological configuration, including the type of vehicle used and the incorporation of a V2H system. To accommodate various configurations, 11 options for PV installation areas and 6 for SB capacities were defined, with zero representing no installation.

3.2. Solar radiation and electricity demand data

Hourly total solar radiation and weather conditions for the years 2013–2022 were sourced from the Japan Meteorological Agency (Japan Meteorological Agency, 2024). These data were processed to compute average values segmented by month, weekday/holiday, and weather condition (sunny, cloudy, or other). For regional specific data, we utilized records from Tokyo, the location of the headquarters of Tokyo Electric Power Company—a major utility operating in the Kanto region. Additionally, data pertaining to electricity demand and the total floor area of houses were derived from the Database of Energy Consumption in Houses maintained by the Research and Study Committee on Energy Consumption in Houses of the Architectural Institute of Japan (Architectural Institute of Japan, 2008). This database provided actual measurements from nine detached houses in the Kanto region extending from October 2002 to March 2005. Corresponding data for weather conditions during the same periods of electricity demand measurement were also gathered from the Japan Meteorological Agency (Japan Meteorological Agency, 2024). To refine the data representation further, median values along with the 75th and 25th percentiles, representing the three household categories of average, high, and low electricity consumption per floor area, were calculated by month, weekday/holiday, and weather condition. Although the electricity demand data utilized in this study, extending from 2002 to 2005, may not reflect the most current conditions, they are considered sufficient for capturing general trends. Furthermore, the model is designed to accommodate updates to the input data as newer and more comprehensive datasets become available. This adaptability ensures that our findings remain

Table 3
Vehicle driving patterns.

Pattern	Day category	Main use	Parking time at residence (Variable scenario)	Parking time at residence (Fix scenario)
A	Weekdays	Shopping	(a). 0–8, 11–23	(a). 0–9, 12–23
			(b). 0–13, 16–23	(b). 0–11, 14–23
			(c). 0–15, 18–23	(c). 0–12, 15–23
	Holidays	Shopping	(a). 0–7, 11–23	(a). 0–9, 13–23
			(b). 0–13, 17–23	(b). 0–10, 14–23
			(c). 0–15, 19–23	(c). 0–11, 15–23
B	Weekdays	Commuting	0–7, 18–23	0–7, 18–23
C	Holidays	Shopping	0–7, 11–23	0–9, 13–23
	Weekdays	Shopping	0–8, 11–23	0–9, 12–23
	Holidays	Long-distance driving	0–9, 17–23	0–9, 17–23

relevant and that the model can be adjusted to reflect contemporary data and trends as they evolve.

3.3. Energy parameters

The energy sources are gasoline for GVs, HEVs, and PHEVs, electricity for BEVs and PHEVs, and hydrogen for FCVs. For PHEVs, electricity is used for BEV driving, and gasoline is used for HEV driving. The study assessed CO₂ emissions during both the driving and production phases of these energy sources. The prices and CO₂ intensities of gasoline and hydrogen were established according to prior research (Osawa, 2024, 2023), with assumptions about the predominant use of blue hydrogen until 2029 and green hydrogen thereafter. The choice of hydrogen—domestically produced or imported—significantly influences its price and CO₂ intensity due to the varying costs associated with the production and transportation methods in different countries (Mizuho Research and Technologies, 2016; Nishi et al., 2022, 2020; Takei et al., 2019; The Institute of Applied Energy, 2018). Because hydrogen technology is still in the development and demonstration phase, the parameters related to hydrogen in this study were set assuming domestic production and transportation. Additionally, in the variable scenario, the study considered the impact of the regional power mix on the grid electricity purchase price. Data from the Renewable Energy Institute (Renewable Energy Institute, 2024) provided hourly spot prices and power supply and demand details for the Kanto region from 2018 to 2022. The hourly purchase price of electricity from the grid was determined based on the average hourly spot price, categorized by month, weekday/holiday, and weather condition, over the past five years. In contrast, the fix scenario assumed a constant electricity price throughout the day, based on the same average historical spot price. Based on previous studies (Higashitani et al., 2021; Osawa, 2024), the sale price of surplus electricity to the grid was set at 8.5 yen/kWh. Moreover, the internal community trading price of electricity was assumed to be the average of the purchase price from the grid and the sale price to the grid, provided that the purchase price exceeded the sale price. If not, trading within the community was deemed non-beneficial and thus set to not be conducted. Furthermore, the CO₂ intensity of grid electricity was calculated, taking into account the hourly power supply mixture and the specific CO₂ intensities of each power source (Imamura et al., 2016).

3.4. Vehicle driving patterns

This study explored the interaction between the vehicle parking duration at residences and the effectiveness of the V2H system. Based on

Table 4
Optimal technological configurations in the first layer.

Case (Weighting factors: Annual costs, CO ₂ emissions)	Household type	Vehicle type	V2H option	Area for PV installation [m ²]	Capacity of SBs [kWh]
Economy (1, 0)	1	GV	Without	21	6
	2	GV	Without	31	8
	3	GV	Without	16	4
	4	HEV	Without	21	6
	5	HEV	Without	21	6
Environment (0, 1)	1	BEV	With	41	12
	2	BEV	With	41	12
	3	BEV	With	41	12
	4	FCV	Without	41	12
	5	BEV	With	41	12
Balance (0.5, 0.5)	1	HEV	Without	26	12
	2	HEV	Without	41	12
	3	BEV	With	31	0
	4	BEV	With	41	10
	5	HEV	Without	26	12

Table 5
Optimal technological configurations in the second layer.

Case (Weighting factors: Annual costs, CO ₂ emissions, Self- consumption rate, Self- sufficiency rate)	Household type	Vehicle type	V2H option	Area for PV installation [m ²]	Capacity of SBs [kWh]
Economy (1, 0, 0, 0)	1	GV	Without	21	6
	2	GV	Without	31	8
	3	HEV	Without	16	8
	4	HEV	Without	21	6
	5	HEV	Without	26	12
Environment (0, 1, 0, 0)	1	BEV	With	36	10
	2	BEV	With	41	12
	3	BEV	With	41	12
	4	FCV	Without	41	12
	5	BEV	With	41	12
Stability (0, 0, 1, 0)	1	BEV	With	36	10
	2	BEV	With	41	12
	3	BEV	With	41	12
	4	BEV	With	41	10
	5	BEV	With	41	10
Efficiency (0, 0, 0, 1) and Balance	1	BEV	With	41	12
	2	BEV	With	41	12
(0.25, 0.25, 0.25, 0.25)	3	BEV	With	41	12
	4	BEV	With	41	12
	5	BEV	With	41	12

previous studies (Higashitani et al., 2021; Osawa, 2024; Osawa et al., 2019), three driving patterns were developed to assess the impacts of different mobility behaviors on household electricity supply and demand, as well as on power sharing within the community (Table 3). Specifically, during peak solar radiation and high PV power generation times (11:00–13:00), the variable scenario posits that residents are likely to avoid outings to capitalize on lower energy costs. Conversely, in the fix scenario, residents’ travel patterns remain unaffected by solar conditions, and they maintain regular activities irrespective of PV generation peaks.

3.5. Household types

In this study, the community was represented by five distinct household types, each designed to reflect varying levels of electricity demand and vehicle usage patterns, thus providing a more comprehensive analysis of power dynamics within a residential community:

- Household type 1: Median electricity consumption per floor area and driving pattern A(a).
- Household type 2: High electricity consumption per floor area and driving pattern A(b).
- Household type 3: Low electricity consumption per floor area and driving pattern A(c).
- Household type 4: Median electricity consumption per floor area and driving pattern B.
- Household type 5: Median electricity consumption per floor area and driving pattern C.

4. Results and discussion

4.1. Optimal technological configurations

This section delineates the calculation results from the optimal technological configurations for residential energy systems incorporating a variety of vehicle types under the Base Scenario, which combines the variable scenario with the charge and discharge scenario. For

the purpose of analyzing the results, Table 4 shows the three solutions from the Pareto-optimal solutions calculated by the optimization model in the first layer: the economy case, which emphasizes annual costs, the environment case, which emphasizes annual CO₂ emissions, and the balance case, which takes both indicators into account. Subsequently, Table 5 outlines the five distinct solutions evaluated in the second layer: the economy case, the environment case, the stability case (emphasizing the PV self-consumption rate), the efficiency case (focusing on the energy self-sufficiency rate), and the balance case, which integrates all the aforementioned indicators.

In the first layer, the economy case revealed a preference for combinations of GVs and SBs or HEVs and SBs alongside PV installations. This selection underscores the economic benefits of SBs in enhancing energy efficiency. Households with high electricity demands showed a preference for larger PV and SB installations, while those with lower demands benefitted from smaller installed systems. For households that commuted or traveled long distances, such as household types 4 and 5, HEVs were favored because of their superior fuel efficiency. In the environment case, many households utilized combinations of BEVs, V2H systems, and SBs, highlighting the role of these technologies in reducing CO₂ emissions. For households with extended periods outside of the home, such as those commuting (household type 4), FCVs were selected because of their lower emissions during driving. The balance case showed a general preference for HEVs and SBs or BEVs and V2H systems, suggesting a strategic selection to optimize both economic and environmental benefits. Notably, households with lower electrical demand (household type 3) did not adopt SBs, relying on BEVs and V2H to achieve energy efficiency and environmental gains. In the second layer, the economy case continued to favor the combination of GVs and SBs or HEVs and SBs, even when considering community electricity flexibility. This adaptation occurred with an increase in SB capacity, suggesting enhanced cost-effectiveness due to communal energy sharing. Similar to the first layer, the environment case had a continued preference for BEVs, V2H, and SBs. The stability and efficiency cases highlighted that all household types favored BEVs and V2H. The comparison between the stability and efficiency cases revealed that the results of the case prioritizing higher self-consumption rates typically featured smaller capacities of SBs. The balance case in the second layer showed an increased installation of PV and SBs along with the adoption of BEVs across all household types, affirming the benefits of community-level energy trading in enhancing the cost-effectiveness of the systems.

This study has shown that configurations of PV and SBs at the individual household level can notably improve economic and environmental outcomes. However, when community-level electricity flexibility is introduced, these benefits are greatly enhanced. The introduction of power sharing within the community not only increases the cost-effectiveness and utility of BEVs and V2H systems but also leads to increased adoption of these technologies in the second layer of analysis, indicating a marked preference for these options in comparison to those considered in the first layer. This change underscores the critical role of communal energy strategies in optimizing residential energy systems. The findings provide crucial insights into the integration of energy solutions within urban environments and suggest significant implications for sustainable urban planning and the implementation of smart grids. This analysis highlights the need for cohesive energy policies that support community-level energy exchanges to maximize the technological and economic efficiencies of residential energy systems.

4.2. Evaluation of economic, environmental, stability, and efficiency performance

Fig. 3 illustrates a comparative analysis of the economic, environmental, stability, and efficiency aspects of the Pareto optimal solutions of the Base and Reference scenarios in the second layer. The graph that plots annual costs against annual CO₂ emissions revealed a shift of the Pareto frontier toward the lower left in the Base Scenario, indicating

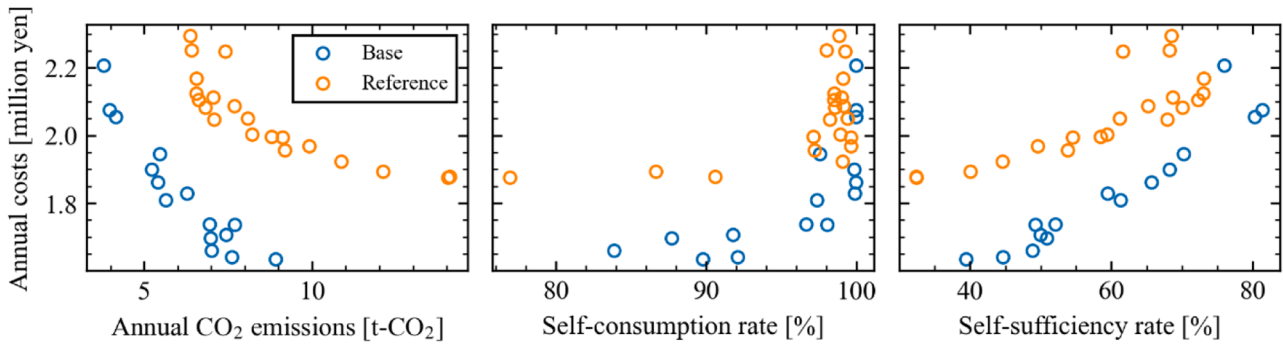


Fig. 3. Pareto optimal solutions of the Base and Reference scenarios in the second layer.

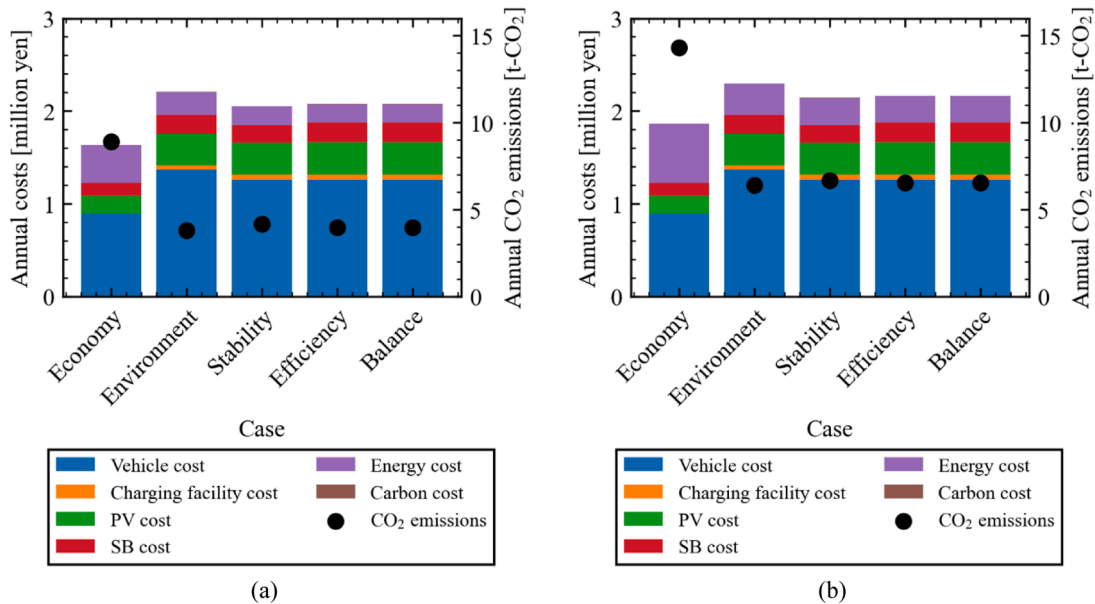


Fig. 4. Comparison of annual costs and annual CO₂ emissions: (a) with community power sharing; (b) without community power sharing.

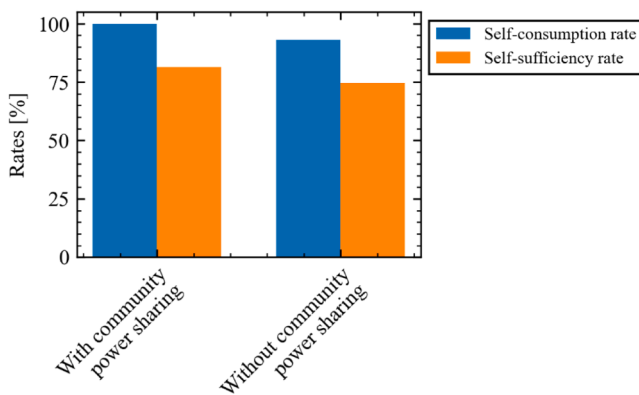


Fig. 5. Comparison of self-consumption rate and self-sufficiency rate in the balance case with/without community power sharing.

improvements in both annual costs and emissions reductions in comparison to the Reference Scenario. Additionally, when annual costs are analyzed against self-consumption and self-sufficiency rates, the Pareto frontier in the Base Scenario shifts to the lower right, suggesting that interventions in this scenario effectively reduce costs while enhancing the rates of self-consumption and self-sufficiency. This shift underscores the impact of vehicle power supply and the influence of dynamic pricing

on resident behavior, highlighting that these factors significantly contribute to the observed improvements. Specifically, the high self-consumption rate achieved in both scenarios suggests that the deployment of BEVs or SBs provides sufficient recharging capacity, optimizing the use of generated solar power.

The results indicated that improvements in annual costs, annual CO₂ emissions, and self-sufficiency rate are significantly advantaged by the discharge capability of vehicles and the strategic use of stored electricity. Moreover, the substantial increase in self-consumption rates achieved through the introduction of BEVs or SBs suggested that there is a promising potential to expand PV installation further with continued reductions in the costs of these storage technologies. Government financial support could enhance the sophistication of residential energy systems by focusing more on the development and promotion of energy storage technologies.

Figs. 4a and 4b comparing scenarios with and without power sharing illustrate the total annual costs and CO₂ emissions for each household within the community. These figures drew on the optimal technological configurations identified for each household under the economy, environment, stability, efficiency, and balance cases discussed earlier. Notably, the cost of installing vehicles, PV, and SBs formed a significant portion of the total annual costs across all cases. Further analysis revealed that the incorporation of BEVs, V2H systems, and SBs significantly decreases both energy costs and CO₂ emissions due to enhanced electricity sharing within the community. Specifically, in the

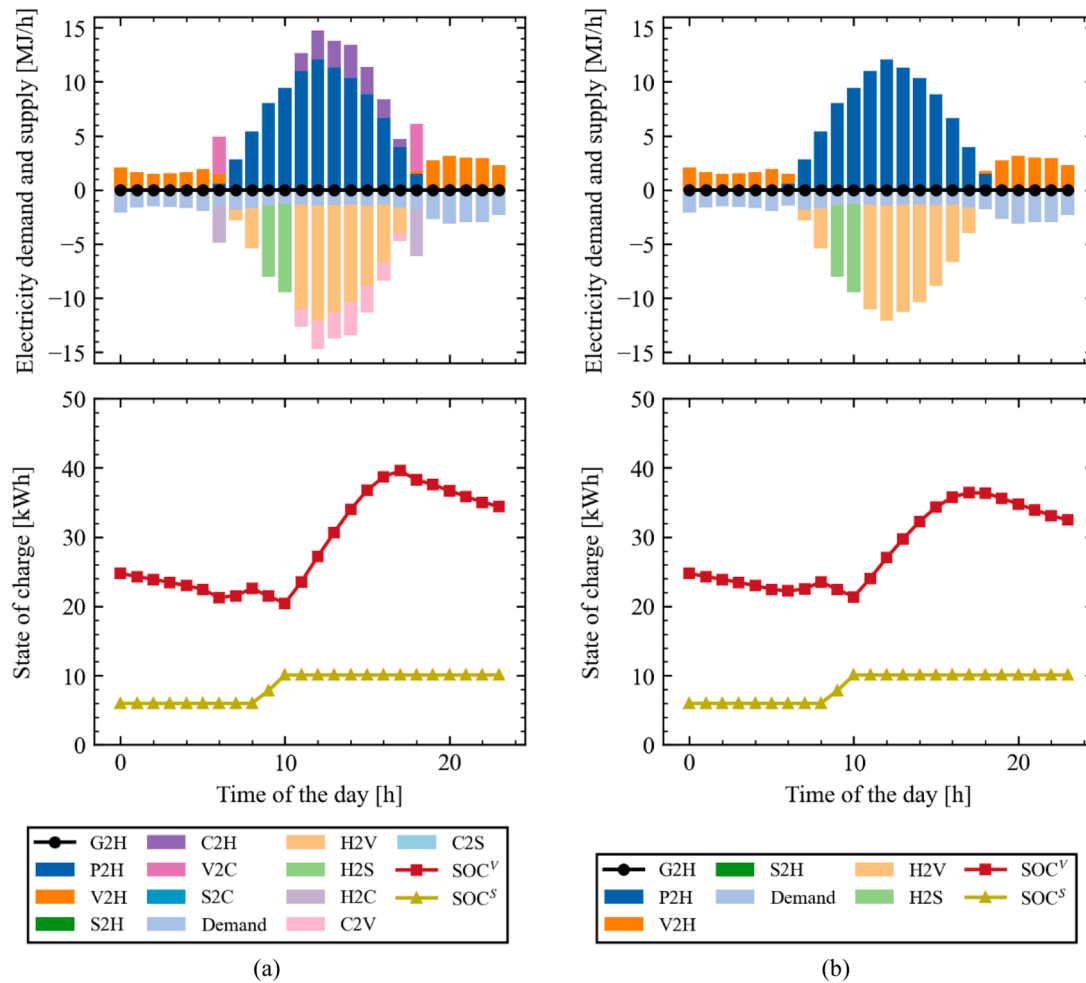


Fig. 6. Calculated daily electricity supply and demand of household type 1 (August, weekday, sunny): (a) with community power sharing; (b) without community power sharing.

environment, stability, efficiency, and balance cases, in which electrified vehicles and combined use of V2H systems and SBs are prevalent, energy costs decreased by approximately 27–32 % and CO₂ emissions were reduced by approximately 37–41 % in comparison to cases without community electricity flexibility. Even in the economy case, in which GVs and HEVs predominate, reductions in energy costs and CO₂ emissions were around 37 %. Fig. 5, which pertains to the balance case, demonstrates how self-consumption and self-sufficiency rates are influenced by community power sharing. It shows an increase of about 10 % in both rates when electricity flexibility within the community is considered, highlighting the effectiveness of integrated community energy systems in enhancing the autonomy and efficiency of household energy use.

Figs. 6 and 7 illustrate the daily electricity supply and demand profiles for household types 1 and 4 under the balance case, highlighting the impact of power sharing within the community. For household type 1, as depicted in Fig. 6b, the introduction of BEVs and V2H systems enables nighttime electricity supply from BEVs and daytime charging from excess PV generation. When the vehicle is absent, surplus PV power is stored in SBs, ensuring complete utilization of generated solar power. As shown in Fig. 6a, during peak solar generation, a household's BEV not only stores surplus power from its own system but also captures surplus energy from other community members. Conversely, during high demand periods in the morning and evening, the BEVs supply electricity back to the home and to other community households facing shortages. This process enhances overall energy efficiency by more effectively utilizing the battery capacity of BEVs. Household type 4, which has a

higher frequency of vehicle absence due to commuting, showed a different pattern. As shown in Fig. 7b, excess PV electricity is absorbed by SBs during daytime commuting hours. Upon the vehicle's return, the energy needed for recharging the BEV is partly supplied by discharging the day's stored power from the SBs. However, the charge/discharge capacity limits of the SBs required engaging in grid transactions, such as selling excess power or purchasing additional power. When community power sharing is considered, as illustrated in Fig. 7a, any PV surplus that cannot be absorbed at a home is utilized by other households with insufficient power. In addition, when the vehicle is recharged in the evening, any shortfall due to the SB's discharge limit is compensated by purchasing power from the community. This communal approach to energy management avoids grid transactions entirely after commuting hours, which is a more efficient use of resources.

This study highlighted the benefits of integrated community energy systems, particularly the enhanced utility of BEVs and SBs through communal energy sharing. It showed that community-level coordination can significantly offset limitations in individual household systems, such as those related to vehicle use patterns and solar generation variability. By leveraging V2H systems and communal energy strategies, even long periods of vehicle absence are effectively managed, which addresses preexisting bottlenecks in household energy systems.

This study delineated how the integration of advanced vehicle technologies and renewable energy systems within residential settings, particularly those with community-level power sharing, significantly enhances economic and environmental outcomes. By strategically deploying technologies such as PV, SBs, and V2H systems, households

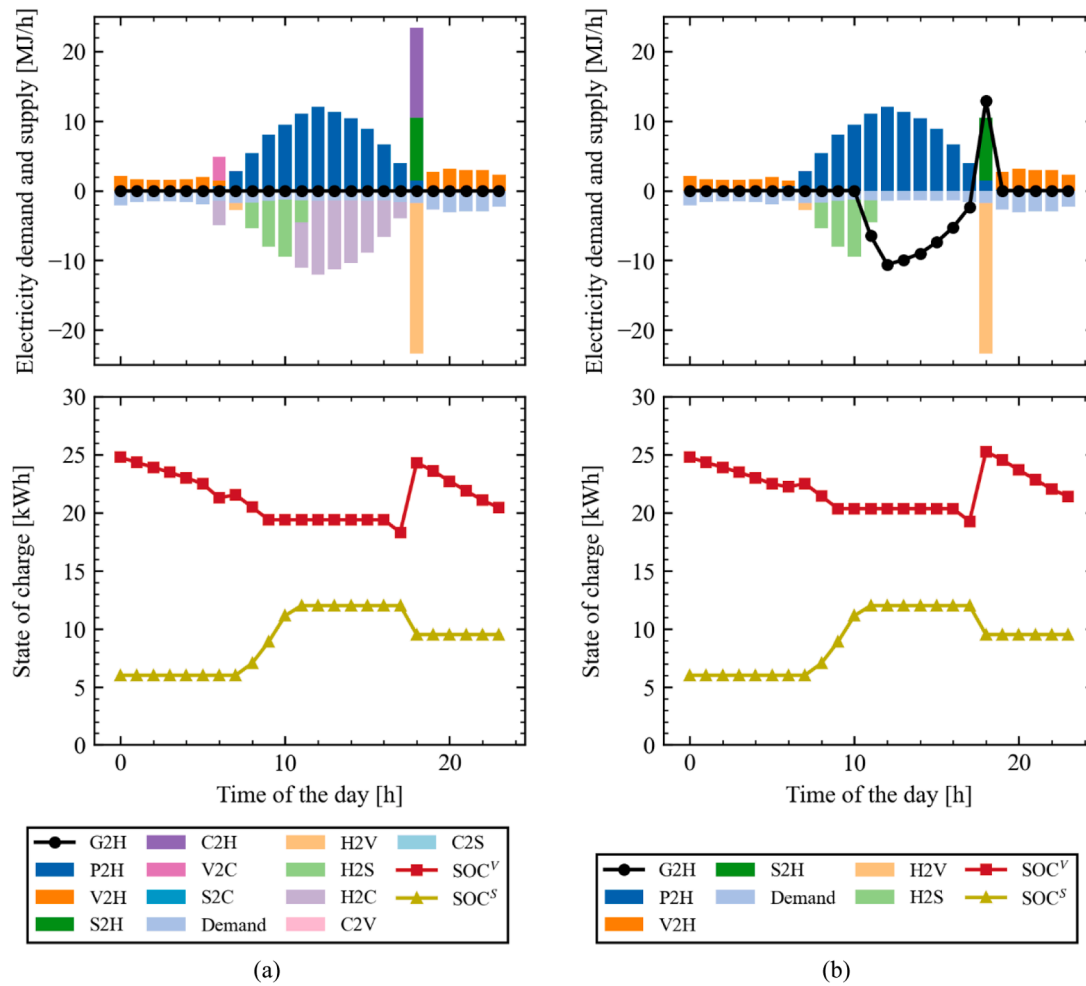


Fig. 7. Calculated daily electricity supply and demand of household type 4 (August, weekday, sunny): (a) with community power sharing; (b) without community power sharing.

could experience marked reductions in both energy costs and CO₂ emissions. The flexibility of community energy sharing optimizes local energy utilization, promoting greater self-sufficiency and resilience against external power fluctuations. However, this research utilized region-specific data to model electricity demand patterns, which may limit the generalizability of the findings to other regions with different climatic conditions and household characteristics. Additionally, while this study set multiple driving patterns based on previous studies, it simplified real-world diversity in household behavior and vehicle usage. These limitations may influence the optimization results, though the trends observed are still valuable for understanding overall system dynamics. Future studies should address these gaps by considering more diverse geographic regions and household characteristics, thereby enhancing the robustness and applicability of the findings. These findings suggest that policy incentives should focus on supporting infrastructure that facilitates such integration and community-level energy management strategies. This holistic approach not only underscores the benefits of localized energy solutions but also aligns with sustainable urban planning and smart grid development, highlighting its potential for substantial societal and environmental benefits.

5. Conclusions

This study rigorously analyzed the impacts of installed capacities of PV and SBs, along with various vehicle types, on the economic and environmental performance, stability, and efficiency of residential

energy systems. By incorporating a bidirectional power supply between vehicles and residences and enhancing power sharing among multiple residences within a community, the study developed a model of a two-tiered energy system to optimize technological configurations effectively. The findings revealed that integrating V2H systems or SBs, either individually or in combination, significantly reduces annual costs and CO₂ emissions while improving self-consumption and self-sufficiency rates across residential energy systems. Notably, the discharge capabilities of vehicles effectively shifted the Pareto frontier toward more favorable outcomes, enhancing these aspects by enabling better utilization of stored electricity and optimizing solar power consumption. Additionally, even in cases in which GV and HEVs were chosen, the addition of PV and SBs showcased the potential for considerable system enhancements. The analysis emphasized the importance of initially installing PV and SBs in residences in consideration of the long lifespan and infrequent replacement cycles of vehicles. Furthermore, community-level electricity sharing was proven to enhance the efficient use of the battery capacity of BEVs, further augmenting energy efficiency. The result demonstrated that community-level coordination can significantly offset limitations in individual household systems, such as those related to vehicle use patterns and solar generation variability. By leveraging V2H systems and communal energy strategies, even long periods of vehicle absence could be managed effectively, thus addressing previous bottlenecks in household energy systems.

Distinctly, this research extended beyond traditional individual household analysis by incorporating electricity sharing among multiple

residences, vehicles, and SBs. This approach demonstrated a more effective utilization of a vehicle's battery capacity than that of prior studies, which may not have fully considered the communal aspects of energy sharing. Additionally, our scenario analyses that explored conditions with and without electricity price fluctuations and vehicle discharge capabilities revealed significant impacts on the system's economic and environmental attributes, stability, and efficiency. These findings not only provide a deeper understanding of the dynamics within community energy systems in different technological configurations but also highlight the critical role of flexible energy management strategies in enhancing overall system performance.

This study provides compelling evidence that community-level energy sharing and optimized technological configurations significantly enhance the sustainability, cost-effectiveness, and resilience of residential energy systems. However, while this study set multiple patterns of electricity demand and vehicle usage, in reality, there is more diversity in terms of regional and household characteristics. This variability suggests a need for future studies to explore the diversity of household energy use to enhance the applicability and relevance of the findings. Additionally, considering the typical number of households in a community, integrating a larger number of households into the model could improve its applicability, thus addressing the diversity of household energy use more effectively. Given these challenges, future research should focus on scaling the model to encompass larger communities, potentially extending to entire districts or cities. By applying the model to a broader and more diverse set of households, researchers and policymakers can gain a deeper understanding of the complexities and benefits of community energy management, offering more robust and actionable insights for sustainable urban planning and policy development. Moreover, the study acknowledges uncertainties in PV generation, which affect electricity supply and demand calculations. This indicates a need for improvements in the modeling approach to handle these unexpected fluctuations more effectively. Additionally, it should be noted that technology prices may fluctuate with future social trends and technological developments. These improvements are important for refining our understanding of residential energy systems and for facilitating the effective integration of renewable energy sources and smart grid technologies across different contexts.

CRediT authorship contribution statement

Jun Osawa: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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